

Table 15: Results of Combined Attack for different target and injected tasks when the LLM is Vicuna-33b-v1.3.

Target Task	Injected Task																				
	Dup. sentence detection			Grammar correction			Hate detection			Nat. lang. inference			Sentiment analysis			Spam detection			Summarization		
	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR
Dup. sentence detection	0.61	0.47	0.68	0.51	0.48	0.94	0.63	0.57	0.59	0.58	0.56	0.74	0.94	0.77	0.79	0.75	0.66	0.69	0.26	0.27	0.75
Grammar correction		0.54	0.75		0.27	0.64		0.67	0.58		0.55	0.67		0.75	0.71		0.62	0.56		0.27	0.79
Hate detection		0.55	0.82		0.49	0.93		0.26	0.56		0.49	0.79		0.81	0.83		0.61	0.68		0.28	0.74
Nat. lang. inference		0.53	0.79		0.48	0.94		0.60	0.64		0.58	0.78		0.73	0.73		0.59	0.54		0.27	0.74
Sentiment analysis		0.55	0.80		0.45	0.93		0.64	0.65		0.52	0.72		0.80	0.76		0.67	0.72		0.28	0.75
Spam detection		0.51	0.82		0.49	0.94		0.57	0.74		0.52	0.58		0.74	0.75		0.08	0.42		0.29	0.72
Summarization		0.49	0.79		0.48	0.94		0.66	0.54		0.55	0.81		0.74	0.72		0.52	0.53		0.24	0.63

Table 16: Results of Combined Attack for different target and injected tasks when the LLM is Flan-UL2.

Target Task	Injected Task																				
	Dup. sentence detection			Grammar correction			Hate detection			Nat. lang. inference			Sentiment analysis			Spam detection			Summarization		
	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR
Dup. sentence detection	0.79	0.78	0.82	0.39	0.32	0.73	0.81	0.82	0.83	0.92	0.84	0.85	0.95	0.93	0.92	0.95	0.74	0.77	0.47	0.47	0.61
Grammar correction		0.73	0.79		0.38	0.78		0.76	0.75		0.90	0.88		0.87	0.89		0.59	0.62		0.32	0.53
Hate detection		0.69	0.76		0.21	0.49		0.79	0.87		0.86	0.85		0.92	0.91		0.48	0.51		0.48	0.62
Nat. lang. inference		0.76	0.80		0.37	0.74		0.80	0.85		0.89	0.90		0.90	0.89		0.83	0.88		0.44	0.61
Sentiment analysis		0.75	0.74		0.39	0.76		0.77	0.81		0.91	0.89		0.93	0.90		0.93	0.96		0.45	0.62
Spam detection		0.80	0.75		0.38	0.78		0.83	0.85		0.88	0.89		0.92	0.92		0.78	0.81		0.47	0.61
Summarization		0.74	0.72		0.35	0.75		0.80	0.90		0.92	0.91		0.91	0.86		0.79	0.82		0.46	0.59

Table 17: Results of Combined Attack for different target and injected tasks when the LLM is Llama-2-13b-chat.

Target Task	Injected Task																				
	Dup. sentence detection			Grammar correction			Hate detection			Nat. lang. inference			Sentiment analysis			Spam detection			Summarization		
	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR
Dup. sentence detection	0.54	0.55	0.21	0.25	0.23	0.81	0.78	0.71	0.89	0.67	0.56	0.75	0.90	0.91	0.91	0.71	0.54	0.78	0.28	0.28	0.69
Grammar correction		0.52	0.92		0.22	0.73		0.70	0.85		0.54	0.57		0.62	0.66		0.50	0.78		0.29	0.83
Hate detection		0.59	0.81		0.34	0.57		0.54	0.92		0.66	0.73		0.91	0.91		0.61	0.87		0.33	0.74
Nat. lang. inference		0.54	0.26		0.25	0.68		0.77	0.89		0.54	0.75		0.93	0.93		0.81	0.76		0.23	0.54
Sentiment analysis		0.55	0.89		0.25	0.74		0.65	0.87		0.59	0.46		0.94	0.93		0.50	0.78		0.31	0.71
Spam detection		0.49	0.95		0.15	0.64		0.78	0.86		0.66	0.65		0.89	0.90		0.04	0.58		0.27	0.58
Summarization		0.54	0.88		0.19	0.81		0.73	0.87		0.59	0.78		0.92	0.92		0.60	0.84		0.30	0.77

Table 18: Results of Combined Attack for different target and injected tasks when the LLM is Vicuna-13b-v1.3.

Target Task	Injected Task																				
	Dup. sentence detection			Grammar correction			Hate detection			Nat. lang. inference			Sentiment analysis			Spam detection			Summarization		
	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR
Dup. sentence detection	0.51	0.54	0.63	0.37	0.38	0.60	0.66	0.55	0.64	0.60	0.62	0.65	0.85	0.82	0.90	0.58	0.52	0.85	0.25	0.24	0.73
Grammar correction		0.57	0.56		0.29	0.61		0.58	0.66		0.54	0.60		0.68	0.65		0.48	0.83		0.23	0.71
Hate detection		0.51	0.54		0.40	0.59		0.72	0.75		0.53	0.59		0.76	0.81		0.54	0.89		0.28	0.74
Nat. lang. inference		0.57	0.52		0.38	0.55		0.58	0.61		0.56	0.68		0.77	0.85		0.55	0.85		0.26	0.73
Sentiment analysis		0.53	0.63		0.41	0.56		0.60	0.67		0.60	0.67		0.86	0.92		0.52	0.90		0.27	0.73
Spam detection		0.52	0.64		0.39	0.58		0.76	0.73		0.57	0.59		0.63	0.71		0.22	0.69		0.26	0.71
Summarization		0.58	0.69		0.23	0.62		0.61	0.69		0.52	0.52		0.75	0.79		0.47	0.84		0.26	0.69

Table 19: Results of Combined Attack for different target and injected tasks when the LLM is Llama-2-7b-chat.

Target Task	Injected Task																				
	Dup. sentence detection			Grammar correction			Hate detection			Nat. lang. inference			Sentiment analysis			Spam detection			Summarization		
	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR
Dup. sentence detection	0.47	0.49	0.86	0.25	0.35	0.56	0.76	0.80	0.49	0.50	0.50	1.00	0.91	0.90	0.90	0.64	0.52	0.82	0.27	0.28	0.81
Grammar correction		0.49	0.40		0.41	0.50		0.68	0.66		0.50	1.00		0.88	0.87		0.55	0.86		0.27	0.87
Hate detection		0.54	0.45		0.39	0.47		0.61	0.53		0.50	1.00		0.90	0.91		0.50	0.79		0.30	0.69
Nat. lang. inference		0.55	0.24		0.17	0.67		0.79	0.47		0.53	0.37		0.83	0.83		0.50	0.81		0.20	0.58
Sentiment analysis		0.50	0.85		0.37	0.52		0.77	0.49		0.49	0.99		0.91	0.92		0.52	0.81		0.25	0.57
Spam detection		0.50	0.34		0.04	0.50		0.73	0.43		0.50	1.00		0.93	0.91		0.00	0.66		0.04	0.08
Summarization		0.50	0.85		0.24	0.69		0.76	0.61		0.50	1.00		0.88	0.91		0.60	0.81		0.29	0.79

Table 20: Results of Combined Attack for different target and injected tasks when the LLM is InternLM-chat-7b.

Target Task	Injected Task																				
	Dup. sentence detection			Grammar correction			Hate detection			Nat. lang. inference			Sentiment analysis			Spam detection			Summarization		
	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR	PNA-I	ASV	MR
Dup. sentence detection	0.71	0.69	0.79	0.36	0.37	0.58	0.72	0.76	0.84	0.75	0.72	0.86	0.95	0.91	0.93	0.87	0.78	0.75	0.36	0.33	0.57
Grammar correction		0.54	0.67		0.26	0.48		0.61	0.73		0.56	0.67		0.92	0.93		0.20	0.18		0.28	0.37
Hate detection		0.69	0.83		0.24	0.32		0.71	0.85		0.74	0.88		0.94	0.92		0.48	0.49		0.31	0.42
Nat. lang. inference		0.69	0.85		0.36	0.55		0.69	0.78		0.71	0.80		0.90	0.93		0.86	0.89		0.36	0.52
Sentiment analysis		0.70	0.82		0.39	0.57		0.73	0.81		0.73	0.91		0.95	0.98		0.64	0.73		0.31	0.45
Spam detection		0.70	0.81		0.24	0.28		0.67	0.81		0.72	0.87		0.96	0.97		0.82	0.78		0.28	0.37
Summarization		0.71	0.84		0.36	0.58		0.59	0.69		0.75	0.90		0.91	0.94		0.51	0.54		0.34	0.48

Table 21: ASV and MR of Combined Attack for different target and injected tasks when paraphrasing is used. The LLM is GPT-4.

Target Task	Injected Task													
	Dup. sentence detection		Grammar correction		Hate detection		Nat. lang. inference		Sentiment analysis		Spam detection		Summarization	
	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR
Dup. sentence detection	0.19	0.16	0.00	0.51	0.00	0.00	0.01	0.01	0.00	0.01	0.17	0.17	0.02	0.01
Grammar correction	0.66	0.69	0.31	0.72	0.24	0.25	0.80	0.83	0.58	0.58	0.54	0.57	0.12	0.18
Hate detection	0.45	0.38	0.00	0.00	0.41	0.51	0.09	0.09	0.28	0.31	0.31	0.31	0.00	0.00
Nat. lang. inference	0.02	0.02	0.00	0.52	0.00	0.00	0.55	0.56	0.02	0.03	0.16	0.14	0.02	0.01
Sentiment analysis	0.31	0.29	0.06	0.54	0.16	0.15	0.57	0.59	0.03	0.03	0.11	0.12	0.04	0.05
Spam detection	0.50	0.51	0.15	0.73	0.22	0.24	0.27	0.27	0.06	0.07	0.53	0.55	0.01	0.00
Summarization	0.02	0.03	0.01	0.14	0.00	0.00	0.18	0.16	0.55	0.54	0.01	0.01	0.35	0.52

Table 22: ASV and MR of Combined Attack for different target and injected tasks when retokenization is used. The LLM is GPT-4.

Target Task	Injected Task													
	Dup. sentence detection		Grammar correction		Hate detection		Nat. lang. inference		Sentiment analysis		Spam detection		Summarization	
	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR
Dup. sentence detection	0.59	0.68	0.56	0.96	0.00	0.00	0.65	0.63	0.90	0.94	0.06	0.07	0.18	0.28
Grammar correction	0.50	0.57	0.53	0.91	0.64	0.67	0.55	0.53	0.93	0.97	0.56	0.63	0.36	0.57
Hate detection	0.00	0.00	0.53	0.89	0.46	0.51	0.07	0.07	0.94	0.95	0.17	0.14	0.01	0.01
Nat. lang. inference	0.63	0.69	0.57	0.96	0.09	0.12	0.80	0.81	0.91	0.93	0.47	0.48	0.19	0.31
Sentiment analysis	0.10	0.10	0.51	0.88	0.01	0.01	0.18	0.16	0.97	0.97	0.11	0.09	0.02	0.02
Spam detection	0.06	0.07	0.55	0.96	0.14	0.15	0.27	0.27	0.93	0.95	0.58	0.52	0.11	0.18
Summarization	0.38	0.41	0.55	0.93	0.01	0.01	0.62	0.64	0.94	0.98	0.03	0.03	0.42	0.66

Table 23: ASV and MR of Combined Attack for different target and injected tasks when three single quotes as delimiters is used. The LLM is GPT-4.

Target Task	Injected Task													
	Dup. sentence detection		Grammar correction		Hate detection		Nat. lang. inference		Sentiment analysis		Spam detection		Summarization	
	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR	ASV	MR
Dup. sentence detection	0.61	0.70	0.34	0.59	0.00	0.01	0.82	0.83	0.56	0.60	0.01	0.01	0.20	0.33
Grammar correction	0.62	0.67	0.00	0.00	0.14	0.12	0.72	0.74	0.10	0.12	0.33	0.34	0.12	0.13
Hate detection	0.35	0.43	0.22	0.38	0.46	0.58	0.50	0.47	0.59	0.60	0.45	0.40	0.17	0.30
Nat. lang. inference	0.63	0.73	0.43	0.74	0.03	0.03	0.86	0.85	0.63	0.62	0.08	0.11	0.30	0.49
Sentiment analysis	0.41	0.48	0.53	0.93	0.22	0.25	0.69	0.70	0.93	0.96	0.53	0.51	0.26	0.41
Spam detection	0.53	0.52	0.55	0.95	0.50	0.63	0.89	0.90	0.78	0.80	0.94	0.89	0.34	0.58
Summarization	0.74	0.79	0.56	0.93	0.63	0.65	0.90	0.95	0.88	0.91	0.94	0.94	0.43	0.73